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Supplementary material for “Strain-coupled one-dimensional turbulence: a single-line closure for rapid distortion, its measured limits, and the consequence for leading-edge noise”

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This supplement carries two blocks of supporting material referenced from the main text: the validation against the strained-turbulence experiment of ? (§ ??), summarised in the main text’s validation section, and the complete set of intervention categories of the main text’s measured-limits section (§ ??), each carrying the category.variant label introduced there.

S1. Validation against the strained-turbulence experiment of Chen, Meneveau & Katz

The strain-on/strain-off comparison of the main text is model-internal; an external benchmark must confirm that both the rapid-distortion limit and the departure from it are physical. We use the plane-strain experiment of ? (CMK), in which initially near-isotropic turbulence at $R_\lambda \approx 400$ is subjected to a controlled planar straining cycle and the one-dimensional component spectra are compared with rapid-distortion theory. We reproduce the *straining* phase of their cycle (the destraining phases involve a strain reversal outside the present monotonic configuration); its end corresponds to a total deformation $D = \exp(\frac{1}{2} \int S dt) \approx 3$, i.e. $e \approx 2.20$ in the notation of the main text. Unless stated otherwise, the line ensembles here use the configuration of Appendix B of the main text (molecular viscosity $\nu = 10^{-5}$, scale separation $\mathcal{L}/\eta \approx 250$; the value of $\mathcal{L}/\eta$ is stated in that appendix).

In the rapid-distortion limit (eddy events suppressed, inviscid) the model must reproduce the analytic reference of ? that CMK use. Figure ?? shows that it does: the component spectral centroids follow the rigid-translation law, reaching the deformation ratio 3.00 at $e = 2.20$ with the components coincident, and the kinetic energy reaches 1.72 against the exact rapid-distortion amplification 1.69—the residual being the moment closure’s, not the implementation’s (main text, verification section), since with the events suppressed the solver evolves the closure trajectory (1.72) exactly. The truly inviscid setting is required: any nonzero viscosity dissipates the compressed small scales and degrades the rigid translation, an effect we verified directly.

With the eddy events and viscosity active, the full model (SC-ODT, C0.2) is no longer constrained to rigid translation, and the strain drives a measurable anisotropy between the

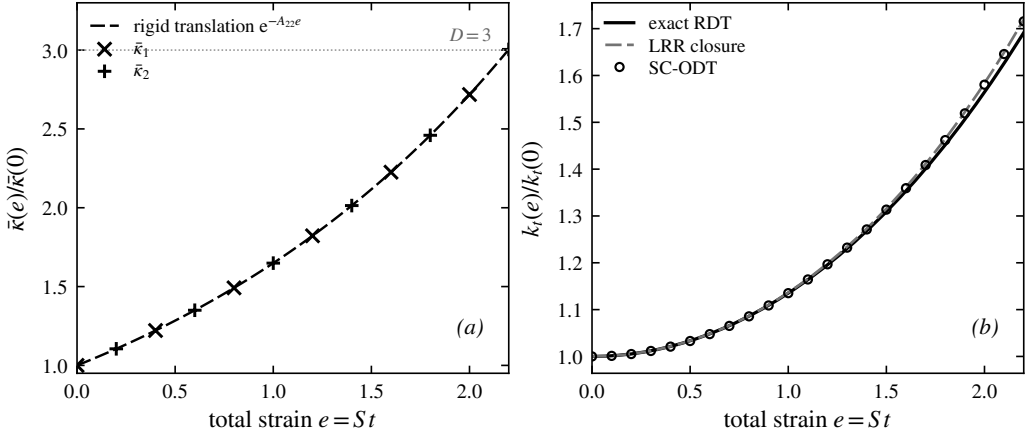


Figure S1. Rapid-distortion limit (eddy events suppressed, inviscid) against the analytic reference of ? used by ?. (a) Component spectral-centroid migration: the streamwise ($\times$) and upwash ($+$) centroids both follow the rigid-translation law and coincide throughout (the markers are interleaved so the two coincident series stay legible), reaching the deformation ratio $D = 3$ at the end of the straining phase. (b) Kinetic-energy amplification: the solver follows the closure trajectory, within 2% of the exact rapid-distortion value at $e = 2.20$.

components. Figure ?? shows the strain-induced migration of the streamwise and upwash component centroids, each measured against the matched no-strain reference (the per-component form of the strain-on/strain-off construction, which cancels the initialisation transient and the eddies' own relaxation): the components diverge progressively under continued strain—the streamwise migration reaching 1.9 at the end of the straining phase against 1.5 for the upwash, both far below the rigid value 3—so the upwash component is held back most. This is the qualitative behaviour CMK report—the strained spectrum departs from the isotropic-translation expectation with the components responding differently—and it is the spectral signature of the broadband distortion analysed in the main text. The comparison is restricted to its qualitative content, for three reasons stated plainly: the band-limited initial spectrum differs from CMK's, so absolute levels are not comparable; the line-resolved one-dimensional spectra and the experiment's transverse/longitudinal projections are not the same projection of the spectrum tensor; and the scale separation achieved here ($\mathcal{L}/\eta \approx 250$) is below the experimental ≈ 930 , whose compressed dissipation range exceeds feasible line resolution. Within these limits the model reproduces the rapid-distortion reference and the qualitative departure mechanism of the benchmark; the quantitative spectral assessment is carried out in the main text against the strained-box DNS, where the projection and the initial spectrum are matched by construction. A direct numerical simulation of the same straining–destraining cycle (?), at a Reynolds number below the experiment's, reports the component variances and the pressure–strain budget through the cycle and compares them with a Launder–Reece–Rodi closure; it is the DNS counterpart against which the present model can be set quantitatively on this configuration. A direct overlay is not attempted here: their observable is the anisotropy through the full straining–destraining cycle, whereas the monotonic straining phase reproduced above does not include the strain reversal, so the comparison is kept at the level of the independent confirmation their DNS provides for the experiment.

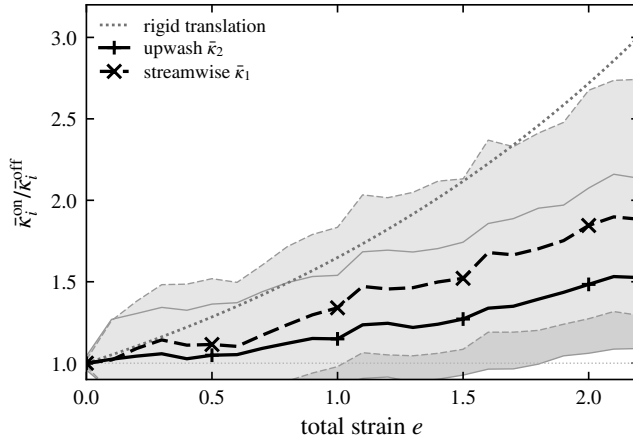


Figure S2. Full model (eddies and viscosity active; ensembles over 128 realisations at $\nu = 10^{-5}$): strain-induced migration of the streamwise and upwash component centroids, each as the geometric-mean ratio of the strained ensemble to the matched no-strain reference, against the rigid-translation law (grey dotted); the shaded bands span ± 1 geometric standard deviation of the ratio across the ensemble. The components diverge under strain—the upwash held back most—reproducing the qualitative departure-from-RDT mechanism of ?. The comparison is qualitative; see text.

S2. The intervention categories and variants

Table ?? records every intervention variant of the measured-limits campaign—its measured fine-scale effect on the transmission diagnostic and the remarks, keyed by the category.variant label—and figure ?? shows the eddy-locked categories on all four diagnostic panels. The variant descriptions, together with the reference configurations C0.1 (standard ODT) and C0.2 (SC-ODT), are collected in Appendix B of the main text. Each case was run at 1024 realisations with seeds paired to SC-ODT (C0.2); effects are median paired same-seed contrasts, with “significant” meaning the 95% bootstrap confidence interval excludes zero. The main text sets out the reading these results establish and retains the two figures that decide it (figures 15 and 16 of the main text: the relaxation clock and the five-way benchmark). The case-by-case record follows.

Two categories fail for instructive reasons. Acceptance gating (C1.1)—rejecting accepted eddies whose post-event state remains anisotropic, at thresholds spanning 45 to 98 per cent rejection—is scale-blind: at mild thresholds the surviving eddy population reproduces the SC-ODT (C0.2) anisotropy budget in every band (the gate self-selects eddies whose kernels were already going to do the relaxing, and the unrelaxed field keeps generating candidates), while at the harshest threshold the model merely drifts towards its eddy-free limit. Restricting the gate to sub-band-scale eddies (C1.2) sharpens the negative result: removing up to 97 per cent of small-eddy events leaves both the fine-scale anisotropy *and* the fine-scale energy unchanged, so no acceptance-gating scheme can control transmission at this Reynolds number. Reweighting the triplet-map images (C2.1; a middle image of relative size $\frac{2}{3}$, which halves the compression of the map’s dominant central channel) moves one-point and energy-containing statistics— $\mathcal{A}_{10}$ falls significantly and the upwash energy fraction with it—but not the fine scales: with the image fractions summing to one, slowing the middle channel necessarily speeds the outer ones, and one outer image maps the energy peak directly into the measurement band.

What works is relaxation applied at the scale of the imbalance. After each accepted eddy’s map and kernels, the same energy-equalising kernel procedure of the main text’s formulation section is applied to each of the three map images—the sub-interval’s own

triplet-map displacement shape as the kernel, coefficients from the sub-interval's own integrals, so that each sub-application conserves the momentum components and the total kinetic energy exactly, with no map applied and no random numbers drawn—and then recursively to 9 and 27 sub-regions. One sub-level (C3.1; $\ell/3$ for an eddy of size ℓ) is inconsequential: for the large eddies that feed the band, $\ell/3$ still sits above it. Two sub-levels (C3.2) produce the first statistically significant fine-scale isotropisation among the categories ($\mathcal{A}_{hi}$ from 1.43 to 1.27 at $e = 1$, significant in both bands), and three sub-levels (C3.3) extend the effect to deep strain: the sub-levels that work, $\ell/9$ and $\ell/27$, are exactly those that reach the measured band. Band energies move by only a few per cent throughout, the upwash energy fraction and the intermittency tails are unchanged, and even at $e = 0$, before any strain, the fine-scale imbalance is pulled towards component equality ($\mathcal{A}_{hi}$ from 0.83 to 0.95).

Four further variants instantiate distinct candidate explanations for the one remaining limitation, the fading of the gains beyond $e \approx 3$. Tying the recursion depth to the eddy size (C3.4; subdividing while the sub-interval remains above a near-dissipative floor, so that large eddies do proportionally more sub-scale work) gives the gains of depth with no reversal at the deepest strain: both bands significant at $e = 1$ and the effect retaining its sign through $e = 3$. Iterating the two-level pass three times within each event (C3.5) answers the amplitude hypothesis in the negative: a stronger early isotropiser that overshoots at deep strain. Spatially delocalising the relaxation—after each eddy, applying the kernel isotropisation in N randomly placed intervals of the same size, overlaps allowed—tests whether the *location* of the events is the limiter: $N = 1$ (C4.1) is inconsequential, and $N = 3$ (C4.2) is significant in both bands at $e = 1$ but acts mainly at the energy-containing scales, since the sampled intervals inherit the eddy-size distribution and full-interval isotropisation acts at that scale. Stacking the two ideas (C4.3)—the adaptive hierarchy inside each sampled interval—is the strongest early isotropiser among the categories ($\mathcal{A}_{hi}$ from 1.43 to 1.26 at $e = 1$) and delivers the only significant deep-strain effect, in the energy-containing band at $e = 3.9$; yet at the fine scales and deep strain it overshoots the depth-only variant significantly. For every scheme locked to the eddy clock, depth sets where the relaxation acts and is necessary; dosage and delocalisation raise the early gains in proportion to the total relaxation delivered; and no eddy-locked combination prevents the fine-scale fade beyond $e \approx 3$.

S3. The relaxation clock in the strained spectra

Figure ?? shows the spectral counterpart of the transmission-diagnostic results of the main text, on the 1024-realisation strained line spectrum ensembles at both rapidities (Appendix B of the main text): the size-capped relaxation clock (C5.2) roughly halves the distance between the model's strained line spectra and the exact rapid-distortion reference family, and—unlike SC-ODT (C0.2)—holds that distance *constant in strain*. A frozen isotropic spectrum is included for reference; it stands for any calculation that feeds a frozen, undistorted inflow spectrum, which cannot represent the strained anisotropy at all, so its distance from the reference family grows to 45–51% by $e = 2$. The component-ratio panel localises the improvement: exact rapid distortion drives the fine scales towards component equality, SC-ODT (C0.2) over-transmits the anisotropy there, and the clock (C5.2) moves the measured ratio towards the linear reference precisely below its size cap ℓ^* .

label	fine-scale effect	remarks
<i>C1 acceptance gating</i>		
C1.1	null at all thresholds; harshest drifts to eddy-free limit	gate is scale-blind: modulates eddy rate, not the scale allocation
C1.2	null, including band energy	fine scales are fed by large-eddy maps directly
<i>C2 map-image geometry</i>		
C2.1	null (one-point statistics move: $\mathcal{A}_{\text{lo}}$ 2.85 $\rightarrow$ 2.50 at $e = 3.9$)	outer images feed the band in one step; channels trade off
<i>C3 sub-scale relaxation kernels (depth)</i>		
C3.1	null within confidence intervals	$\ell/3$ sits above the measured band
C3.2	$\mathcal{A}_{\text{hi}}$: 1.43 $\rightarrow$ 1.27 at $e = 1$ (significant, both bands)	first significant fine-scale isotropisation
C3.3	significant at $e = 1$ and $e = 3$ (−12% paired)	deepest levels reach the band
C3.4	significant in both bands at $e = 1$; no late-strain reversal	best of the categories: gains without the deep-strain overshoot
C3.5	strongest at $e = 1$ (1.29); overshoots at deep strain	per-event amplitude is not the limiter
<i>C4 concurrent relaxation</i>		
C4.1	null	location alone is not the lever
C4.2	significant in both bands at $e = 1$, strongest at the energy-containing scales; fade persists	the sampled intervals act at eddy scale, so delocalising the events does not reach the fine scales
C4.3	strongest early effect of the eddy-locked categories ($\mathcal{A}_{\text{hi}}$ 1.43 $\rightarrow$ 1.26); low band significant even at $e = 3.9$; fine scales overshoot depth-only at deep strain	dosage helps early and at large scales, and harms deep-strain fine scales wherever it is applied
<i>C5 independent relaxation clock</i>		
C5.1	significant in both bands at every strain, growing with strain—the fade broken—but the one-point anisotropy is suppressed ($u_2^2/2k_t$ 0.58 $\rightarrow$ 0.41 at $e = 3.9$)	event-locked timing was the deep-strain limiter; a scale-blind event sequence fights the strain itself
C5.2	fine scales significant at every strain with no fade ($\mathcal{A}_{\text{hi}}$ held at 1.30–1.44); one-point trajectory intact for $e \leq 2$, mildly reduced beyond	timing + scale selectivity: the working form of the mechanism
C5.3	fine scales significant at every strain, held below the matched-rate clock ($\mathcal{A}_{\text{hi}}$ 1.15–1.40); one-point anisotropy reduced further ($u_2^2/2k_t$ to 0.48 at $e = 3.9$ against SC-ODT’s 0.58)	the clock rate is a control: more relaxation holds the fine scales harder, at a proportionate one-point cost

Table S1. The intervention categories on the transmission diagnostic, labelled by category.variant; variant descriptions and the reference configurations C0.1 (standard ODT) and C0.2 (SC-ODT) are in Appendix B of the main text. Each case: 1024 paired-seed realisations; effects are median paired same-seed contrasts, “significant” meaning the 95% bootstrap confidence interval excludes zero. The size-capped clock **C5.2** (the clock-relaxed ODT of the main text), with C0.1 and C0.2, forms the five-way comparison.

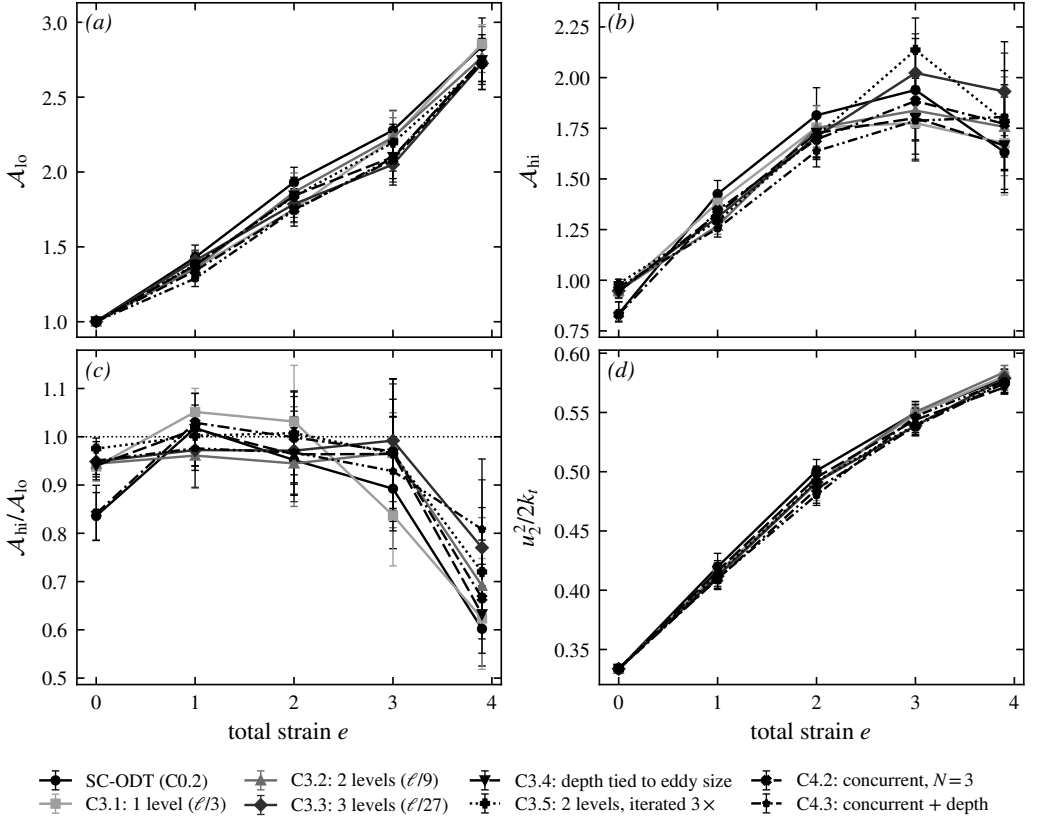


Figure S3. Scale-local relaxation on the transmission diagnostic: (a) energy-containing-band imbalance $\mathcal{A}_{lo}$, (b) fine-scale-band imbalance $\mathcal{A}_{hi}$, (c) transmission $\mathcal{A}_{hi}/\mathcal{A}_{lo}$, (d) upwash energy fraction, against total strain: the SC-ODT reference (C0.2), sub-scale relaxation kernels at fixed depth (C3.1–C3.3), depth tied to eddy size (C3.4), the per-event iterated variant (C3.5), concurrent relaxation (C4.2, $N = 3$ randomly placed same-size intervals per event), and the stacked variant (C4.3, concurrent intervals each running the adaptive hierarchy). Medians over 1024 paired-seed realisations; error bars are the 95% bootstrap confidence intervals of the median.

S4. Operational steps of the leading-edge-noise chain

This section collects, in operational order, the steps of Part III of the main text, from strained line ensembles to radiated level; every step restates the main text.

1. *Inflow representation.* For each total strain e and strain-rate parameter, fit the exact-RDT-distorted von Kármán family to the median component line spectra of the 1024-realisation ensemble—SC-ODT and, identically, the clock-relaxed variant—jointly in the upwash and transverse components, with the pre-distortion amplitude A_0 and energy wavenumber κ_e the only free parameters. The rms log-residual of the fit is the distance envelope of the main text, carried through the chain as the input uncertainty.
2. *Two-dimensional upwash spectrum.* Form $\Phi_{ww}(K_x, K_y; e) = A_0 \int \Phi_{22}^{\text{RDT}}(K_x, \kappa_2, K_y; e) d\kappa_2$ by direct integration of the distorted components of the fitted family; the integration is verified against the wavevector-ensemble integrator of the main text (upwash-variance growth reproduced to under 1%).
3. *Level and shape.* The noise kernel is linear in the upwash variance, so the variance ratio $\overline{u_2^2(e)}/\overline{u_2^2(0)}$ is taken from the measured line spectra and applied as the level; the fitted family supplies only the shape.

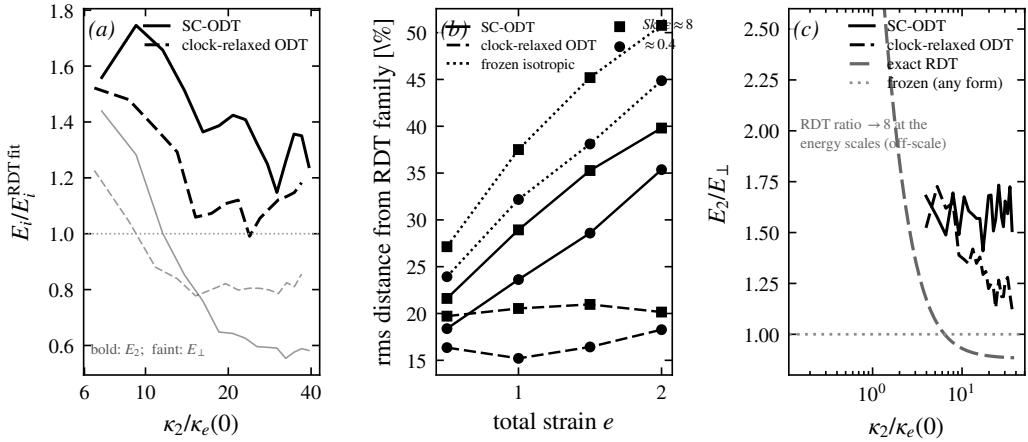


Figure S4. (a) Binned component spectra over their best exact-RDT-family curves at $e = 2$, $Sk_t/\varepsilon \approx 8$ (solid: E_2 ; dashed: $E_\perp$): the SC-ODT (C0.2) misallocation (upwash high, transverse low) and its reduction by the clock (C5.2). (b) rms log-distance from the exact-RDT family against total strain at both rapidities: frozen isotropic representation (gold), SC-ODT (C0.2, black), size-capped clock (C5.2, red); the clock's distance is strain-independent. (c) Component ratio $E_2/E_\perp(\kappa_2)$ at $e = 1$, rapid: measured SC-ODT (C0.2) and clock (C5.2) ensembles against the exact-RDT reference; any frozen form predicts the ratio 1 identically.

4. *Geometry-free ratio.* Form $\mathcal{K}(K_x) = \pi \Phi_{ww}(K_x, 0)^2 / \int \Phi_{ww}(K_x, K_y) dK_y$ for the distorted and for the frozen ($e = 0$) input; $\Delta \text{SPL}(K_x) = 10 \log_{10}[\mathcal{K}^{\text{distorted}}/\mathcal{K}^{\text{frozen}}]$, from which aerofoil geometry, response and observer cancel.
5. *Absolute prediction.* At the reference configuration, physical units enter through two matching conditions only—the integral scale through $\kappa_e = 0.7468/\mathcal{L}$ and the $e = 0$ upwash variance—and the far-field spectrum is assembled from the compact lift-dipole form of ? with the exact Sears response (?); the frozen baseline is then the standard Amiet–von Kármán prediction with no recalibration.
6. *Validation chain.* Against the published measurements, all four inflows (frozen von Kármán, frozen Liepmann, SC-ODT, clock-relaxed ODT) pass through the identical non-compact response, with the Gershfeld thickness admittance applied to all four chains on the NACA0012 cases and the effective strain taken as the potential-flow stagnation-line accumulation truncated at the eddy scale.

S5. Verification of the implementation

The rapid-distortion checks of the main text verify the *formulation*; this section verifies the *implementation*, in which the operator is not prescribed but reconstructed at every substep: the line Reynolds stress is formed from the resolved field, the closure supplies $\Pi_{ij}^{(r)}$, the Lyapunov equation of the main text is solved for B_{ij} , and $\mathcal{A}_{ij} = -A_{ij} + B_{ij}$ is applied pointwise. To isolate this chain the solver is run in the rapid-distortion limit—plane strain, eddy events suppressed, zero molecular diffusion—from a statistically isotropic resolved initial field, and the component energies are evaluated as length-weighted line averages. With the events inactive the line's component-energy evolution must coincide with the closed moment equation across the whole strain range, not merely at onset.

Figure ?? shows that it does: the solver output lies on the LRR trajectory over $0 \leq e \leq 4$ to within 10^{-4} in each component fraction, so the per-substep stress reconstruction, Lyapunov solve and operator assembly are correct at every state of anisotropy. The comparison must

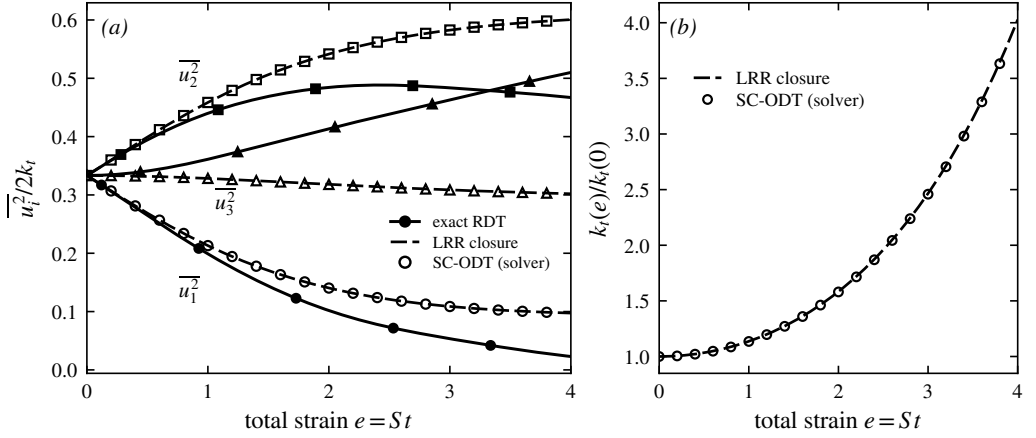


Figure S5. Verification of the SC-ODT implementation under plane strain, eddy events suppressed and zero molecular diffusion. (a) Component energy fractions; (b) kinetic-energy amplification $k_t(e)/k_t(0)$. Open symbols: the SC-ODT line solver (length-weighted line averages), by component ($\circ \overline{u_1^2}$, $\square \overline{u_2^2}$, $\triangle \overline{u_3^2}$); dashed: the LRR closure integrated as a moment equation; solid with filled symbols: exact rapid-distortion theory. The solver reproduces the closure to 10^{-4} in the fractions and 0.3% in the energy; the offset from the exact solution is the closure error quantified in the main text, as it must be.

be read correctly: the solver reproduces the *closure*, and it should—with the events and diffusion suppressed the line carries no information beyond the single-point statistics the closure itself uses. The offset from the exact solution is the single-point closure error quantified in the main text’s finite-strain comparison, not an implementation error; agreement with the exact solution here would indicate a fault. The fractions test only the anisotropy; the kinetic energy, which the traceless rapid term cannot change directly and which grows through the production integral $d \ln k_t / de = f_{22} - f_{11}$, provides the independent magnitude check, and the solver reproduces the LRR amplification to within 0.3%. The test requires a resolved spectral initialisation (a white-noise start places energy at the grid scale, where mesh-adaption interpolation damps it) and a strain-resolved timestep bound in the inviscid limit; both effects act nearly isotropically, so the energy panel is what distinguishes them.

S6. The strained ODT line

Figure ?? is a space–time view of a single realisation of the line under the three model variants of the main text, advanced from identical initial fields (matched seed and spectral initial condition, so the three differ only by the model). Standard ODT (strain off) keeps its full width; SC-ODT and clock-relaxed ODT compress under the plane strain, the physical line half-width (black envelope) falling by a factor of about seven over $e = 0\text{--}4$. The upwash velocity is normalised at each instant so the eddy structure stays visible as the fluctuations decay; amplitudes are therefore not comparable across strain, only the structure and the geometric compression. The clock-relaxed strip retains finer structure at high strain than SC-ODT—the single-realisation signature of the fine-scale hold that the 1024-realisation statistics of § ?? quantify.

The difference between SC-ODT and the clock-relaxed model is not apparent in a single realisation of the velocity itself—the two diverge into distinct realisations—but it shows in the small-scale *structure*. Figure ?? isolates it: the high-pass (small-scale) upwash field over the high-strain window $e = 2.5\text{--}4$, in the stretched coordinate $y/\text{half-width}$ so the

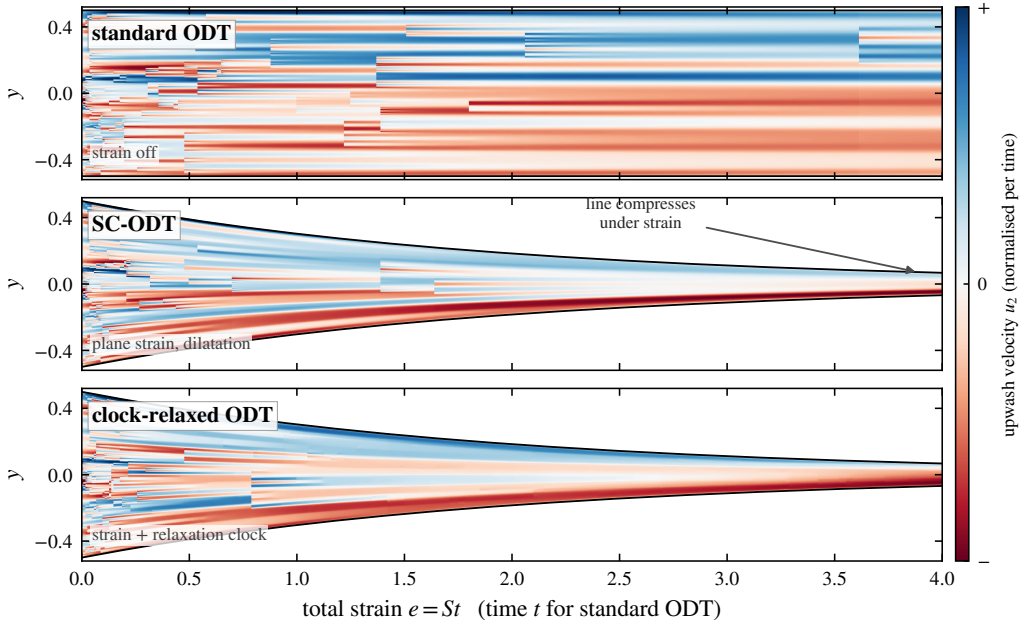


Figure S6. Space–time evolution of the ODT line, upwash velocity u_2 (normalised at each instant; blue positive, red negative, colour intensity the magnitude), for standard ODT, SC-ODT and clock-relaxed ODT, advanced from matched initial fields. Time runs left to right as the total strain $e = St$ (elapsed time for the unstrained standard-ODT case); the black envelope is the physical line half-width. Standard ODT keeps its full width; SC-ODT and clock-relaxed ODT compress under the plane strain, the clock-relaxed line retaining finer structure at high strain.

compressed line fills the frame. SC-ODT smooths to a few coarse bands as the fine scales relax away, whereas the clock-relaxed line stays grainy—its sub-scale relaxation events keep fine-scale structure alive, the single-realisation counterpart of the sustained fine-scale imbalance seen in the ensemble statistics of § ??.

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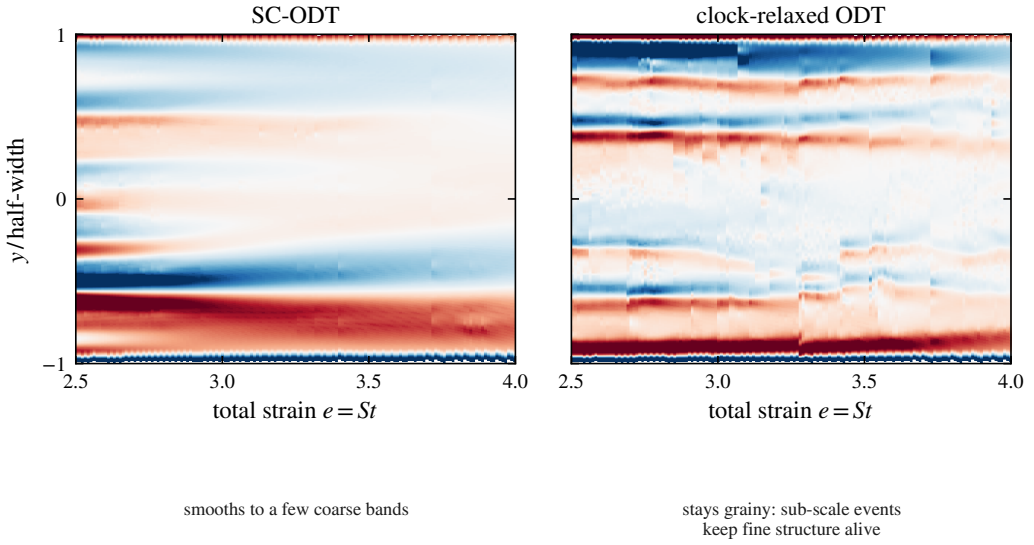


Figure S7. Small-scale (high-pass) upwash structure of a single ODT line in the high-strain window, $e = 2.5$ – 4 , in the stretched coordinate $y/\text{half-width}$: SC-ODT (left) against clock-relaxed ODT (right); blue positive, red negative, colour intensity the magnitude. SC-ODT smooths to a few coarse bands; the clock-relaxed line stays grainy, its sub-scale relaxation events sustaining fine-scale structure (the vertical striations are individual events). Amplitudes are normalised and the two are different realisations, so the contrast is the clock's structural signature, not a pointwise difference.